

Creation Of CV Using LLMs

HPPCS01

Project Abstract

Here the project develops a local, privacy-conscious CV creation and tailoring system using two open-source large language models through Ollama. This system accepts unstructured resumes in PDF, DOCX, or legacy DOC format and converts them into structured candidate facts. A job description is parsed into requirements, responsibilities, and ATS-relevant keywords. Gemma 3 1B is used for information extraction, job-description parsing, and ATS-oriented resume generation. Llama 3.2 1B independently reviews the generated resume for factual consistency, keyword alignment, and formatting quality. Deterministic Python checks provide an additional grounding layer to detect missing source facts, while one controlled regeneration is allowed when the review identifies material issues. Here final resume is generated as DOCX and can also be converted to PDF when LibreOffice is available. The system persists structured job description data and produces evaluation information alongside the final CV. The design focuses on source-grounded rewriting rather than inventing qualifications, technologies, metrics, or achievements, making the generated CV more reliable and suitable for ATS-oriented job applications.

1. Project Introduction

Recruiters commonly receive resumes in different document formats and need to compare them with job-specific requirements. Manual tailoring is repetitive and can cause important experience or ATS keywords to be missed. This project automates that workflow while keeping the candidate document as the source of truth. The approach demonstrates practical use of local LLMs for document understanding, controlled generation, and quality evaluation.

2. Project Problem Statement

The system transforms an unstructured candidate profile into a professional, job-targeted CV without introducing unsupported information. It extracts relevant facts, understands a target job description, identifies matching keywords, generates an ATS-friendly resume, and independently validates the result. It also handles PDF, DOCX, and legacy DOC resume formats.

3. Project Objectives

- Extract candidate name, contact details, education, experience, skills, achievements, and projects.
- Parse a job description into job title, requirements, responsibilities, and ATS keywords.
- Tailor resume wording and ordering to the target role while preserving source-grounded facts.
- Use two independent LLM roles for generation and quality review.
- Apply deterministic grounding and ATS checks with one controlled regeneration when required.
- Produce final DOCX output and PDF output when LibreOffice is available.

4. Methodology Used

The implementation uses Python, Ollama, Gemma 3 1B, Llama 3.2 1B, pypdf, python-docx, requests, and LibreOffice. Here the workflow is: resume ingestion → text extraction → Gemma-based candidate fact extraction → Gemma-based job-description parsing → deterministic candidate/keyword matching → Gemma ATS resume generation → Llama independent review → source-coverage validation → one controlled regeneration when necessary → DOCX/PDF generation. Prompts explicitly prohibit invented employers, titles, dates, certifications, technologies, metrics, responsibilities, achievements, or other unsupported claims.

5. System Design and Implementation

The solution uses a single main.py entry point with functions for document ingestion, Ollama communication, JSON parsing, profile normalization, job-description parsing, keyword matching, resume generation, independent review, grounding checks, and document generation. PDF and DOCX files are parsed directly; DOC files are converted to text through headless LibreOffice. Gemma performs extraction and generation, while Llama acts as an independent reviewer. If review or deterministic validation identifies a material issue, the review reasons are sent back to Gemma for one controlled regeneration. The final DOCX is saved and LibreOffice is used to create a PDF when available.

6. Results and Analysis

The implementation addresses the HPPCS01 objective of generating final CVs from unstructured user-profile documents. It preserves the candidate source as the factual authority, calculates deterministic keyword coverage against the parsed job description, and persists the structured JD for auditability. The independent review evaluates overall score, keyword coverage, factual consistency, and format score. This combines LLM based qualitative review with deterministic safeguards. Generated CVs use standard ATS headings and concise professional content, while unsupported job-description skills are not claimed merely because they appear in the vacancy.

7. Conclusion and Future Work

The project demonstrates an end-to-end local LLM workflow for grounded CV creation and job tailoring. Separating generation from review improves control over hallucination and makes evaluation more transparent. Future enhancements could include OCR for scanned PDFs, richer semantic matching beyond simple keyword containment, configurable resume templates, automated regression testing across larger datasets, and additional export formats. These are optional extensions; the current implementation focuses on the required HPPCS01 objective and reproducible local execution.

8. References Used For Project

1. HAAI++ Capstone Project Instructions, HPPCS01 – CV Creation using LLMs.
2. HAAI++ project brief, “CV Creation using LLMs”, covering resume extraction, job-description parsing, tailoring, user review, and evaluation criteria.
3. Ollama local LLM runtime and API documentation used by the implementation.

Acknowledgement

This project was developed as part of the HAAI++ capstone program and applies the supplied project objective and evaluation guidance to a practical local LLM-based CV creation workflow.